

CSE 309 Theory of Automata, Spring 2026
Exercise 9

1. **Non-Deterministic Polynomial Closure.** Show that the class NP is closed under union and concatenation.
2. **Graph Isomorphism.** Call graphs G and H isomorphic if the nodes of G may be reordered so that it is identical to H . Let

$$ISO = \{ \langle G, H \rangle \mid G \text{ and } H \text{ are isomorphic graphs} \}.$$

Show that $ISO \in \text{NP}$.

3. Show that NP is closed under the star operation.
4. **SAT.** Does there exist an assignment to the boolean variables x and y that results in the following expression evaluating to 1?

$$(x \vee y) \wedge (x \vee \bar{y}) \wedge (\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})$$